

Class Test - 01

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uses algorithms to analyze images and videos to identify the objects.

to understand visual data
to detect and recognize objects

Real-World applications

Face Recognition

Self-driving cars.

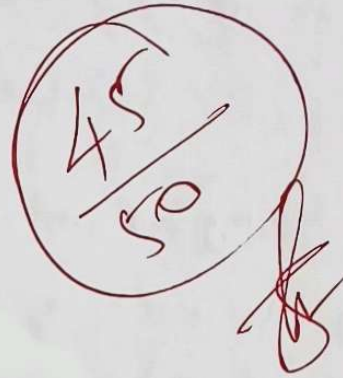


Image analysis, object detection, feature extraction, and Pattern recognition.

Feature extraction: The process of identifying the characteristic from an

image to identify and classify the object and image.

What are the different levels of image processing used:

Low-level image processing: It enhances the quality of the image.

Mid-level image: Image segmentation and feature extraction.

High-level image: object recognition and scene understanding.

Digital image made up of pixels arranged in rows and columns.

Pixel: The smallest unit of digital image

Resolution: Higher resolution gives better image quality.

⑤ It Captures a real-world scene and convert it into a digital image.

a) Role of an image sensor : It Converts lights into electrical signals.

b) Two image acquisition devices:

- Digital Camera
- Scanner.

⑥ How do Sampling and quantization contribute

a) Sampling : It Converts a Continuous image into pixels.

b) Quantization : It Assigns intensity values to pixels.

⑦ It explains how light forms an image on a sensor.

a) Perspective projection : It Converts a 3D scene into a 2D image

b) Role of illumination : It Provides light needed to capture clear image.

⑧ How are image processing and Computer vision related.

a) Image processing : Enhance or modifies the image.

b) Two differences:

Image processing improves image; Computer vision understands image.
Image processing produces enhanced images; Computer vision makes decisions.

is Computer vision applied in real life.

Health care: It detects diseases from medical image

Autonomous vehicles: It detects roads, vehicles and pedestrians.

How do digital image fundamentals supports Computer vision.

Why are pixel important

Pixel stores the image information.

Sampling and resolution:

Higher Sampling and resolution provides more details for accurate image analysis.